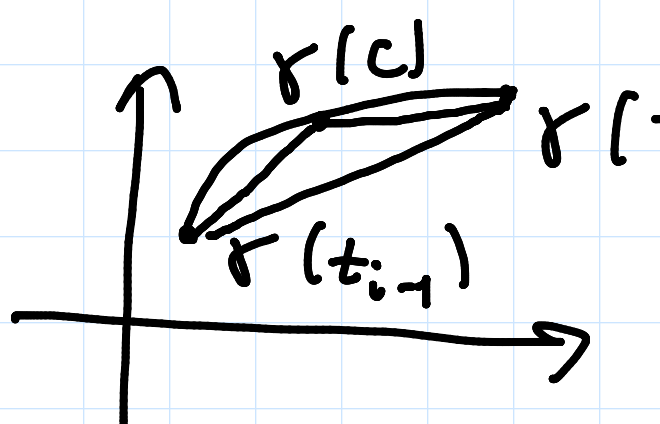


$\textcircled{T} \textcircled{g}$ $\exists \tau: [a, b]$. Если $c \notin \tau$, то добавим c в τ .
 $\Rightarrow \mathcal{D}\tilde{\tau} = \tau \cup \{c\}$, $\int \tilde{\tau} = \tau_1 \cup \tau_2$: $\tau_1: [a, c]$,
 $\tau_2: [c, b]$. $\exists c \in (t_{i-1}, t_i) \Rightarrow$
 $|h_{\tilde{\tau}}| \geq |h_{\tau}|$!

Т.к. $\sqrt{[x(t_i) - x(t_{i-1})]^2 + [y(t_i) - y(t_{i-1})]^2} \leq \sqrt{[x(c) - x(t_{i-1})]^2 + [y(c) - y(t_{i-1})]^2} + \sqrt{[x(t_i) - x(c)]^2 + [y(t_i) - y(c)]^2} \Rightarrow$



$\Rightarrow |h_{\tau}| \leq |h_{\tilde{\tau}}| \leq |h_{\tau_1}| + |h_{\tau_2}| \leq$
 $\leq \ell_{\gamma^*} + \ell_{\tilde{\gamma}} < +\infty \Rightarrow \{\gamma\}$ — суп-м
 $\xrightarrow{\sup} \sup |h_{\tau}| \leq \ell_{\gamma^*} + \ell_{\tilde{\gamma}} \Rightarrow \ell_{\gamma} \leq \ell_{\gamma^*} + \ell_{\tilde{\gamma}}$

$\textcircled{H} + \textcircled{g} \Rightarrow \ell_{\gamma} = \ell_{\gamma^*} + \ell_{\tilde{\gamma}} \Delta$

$\textcircled{T_2}$ Если γ — максим. нр, то γ — суп-м.
 (гос. уст. суп-м).

$\Delta \exists \tau: [a, b]$.

$|h_{\tau}| = \sum_{i=1}^n \sqrt{[x(t_i) - x(t_{i-1})]^2 + [y(t_i) - y(t_{i-1})]^2} =$
 $= \sum_{i=1}^n \sqrt{x'(\xi_i)^2 \Delta t_i + y'(\eta_i)^2 \Delta t_i} \Rightarrow$

$\sum_{i=1}^n \sqrt{m_x^2 + m_y^2} \Delta t_i \leq |h_{\tau}| \leq \sum_{i=1}^n \sqrt{M_x^2 + M_y^2} \Delta t_i$

$m_x = \min_{[a, b]} |x'(t)|$

$m_y = \min_{[a, b]} |y'(t)|$

$$M_x = \max_{[a, b]} |x'(t)|$$

$$M_y = \max_{[a, b]} |y'(t)|$$

$$\Rightarrow \sqrt{m_x^2 + m_y^2} (b-a) \leq |L_\tau| \leq \sqrt{M_x^2 + M_y^2} (b-a)$$

$$\sqrt{m_x^2 + m_y^2} (b-a) \leq \sup_{\tau} |L_\tau| \leq \sqrt{M_x^2 + M_y^2} (b-a)$$

$\parallel$
 L_τ

$$\Rightarrow \gamma - \text{enf} - \bar{u} \text{ нут } \Delta$$

$\textcircled{T_3}$] $\gamma: [a, b] \rightarrow \mathbb{R}^n$ — магкун $\Rightarrow$

φ -эле $\ell_\gamma(t) \in C^1[a, b]$, где

$\ell_\gamma(t)$ — элемент касательной к γ в

т. $\gamma(a)$ го $\gamma(t)$, $t \in [a, b]$

Δ] $t_0, t_0 + \Delta t \in [a, b]$, $\Delta t > 0$

$\textcircled{T_2} \Rightarrow \sqrt{m_x^2 + m_y^2} \Delta t \leq \ell_\gamma(t_0 + \Delta t) - \ell_\gamma(t_0) \leq \sqrt{M_x^2 + M_y^2} \Delta t$

$$\sqrt{m_x^2 + m_y^2} \leq \frac{\ell_\gamma(t_0 + \Delta t) - \ell_\gamma(t_0)}{\Delta t} \leq \sqrt{M_x^2 + M_y^2}$$

$\Delta t \rightarrow 0+0$
 $\downarrow$
 $x'^2(t_0) \quad y'^2(t_0)$

$\Delta t \rightarrow 0+0$
 $\downarrow$
 $x'^2(t_0) \quad y'^2(t_0)$

$$\sqrt{x'^2(t_0) + y'^2(t_0)} \leq \lim_{\Delta t \rightarrow 0+0} \frac{\ell_\gamma(t_0 + \Delta t) - \ell_\gamma(t_0)}{\Delta t} \leq \sqrt{x'^2(t_0) + y'^2(t_0)}$$

$$\Rightarrow \ell'_{\gamma+}(t_0) = \sqrt{x'^2(t_0) + y'^2(t_0)}$$

Если $\Delta t < 0$ — аналогично. Т.к. $t_0 - \forall$

$$\ell'_\gamma(t) = \sqrt{x'^2(t) + y'^2(t)} \Rightarrow \ell_\gamma(t) \in C^1[a, b]$$

$$d\ell(t) = \sqrt{x'^2(t) + y'^2(t)} dt$$

Т₄] $\gamma: [a, b] \rightarrow \mathbb{R}^n$ — регулярная кривая $\Rightarrow$

$$L_\gamma = \int_a^b \sqrt{x'^2(t) + y'^2(t)} dt$$

Δ no $T_3: \ell'_\gamma(t) \in C^0[a, b] \xRightarrow{\text{ФН-1}}$

$$\ell_\gamma(t) = \underbrace{\ell_\gamma(t)}_{\text{в}} - \underbrace{\ell_\gamma(a)}_{\text{в}} = \int_a^t \ell'_\gamma(z) dz$$

Т.к. $\ell_\gamma = \ell_\gamma(b) \Rightarrow$

$$\ell_\gamma = \int_a^b \ell'_\gamma(z) dz = \int_a^b \sqrt{x'^2(t) + y'^2(t)} dt$$

Лемма.] $f \in C^1[a, b]$,

$$] \Gamma_f = \{ (x, y) : y = f(x), x \in [a, b] \}$$

$$\Rightarrow L(\Gamma_f) = \int_a^b \sqrt{1 + f'^2(x)} dx$$

$\Delta \Gamma_f: \gamma(t) = \begin{cases} x = t \\ y = f(t) \end{cases} \quad t \in [a, b]$ — параметризация Γ_f

$$\xRightarrow{T_4} L(\Gamma_f) = \int_a^b \sqrt{1 + f'(x)^2} dx. \quad \Delta$$

Лемма] $f \in C^1[\alpha, \beta]$, $f' > 0$

$$] \Gamma_f = \{ (\varphi, r) : r = f(\varphi), \varphi \in [\alpha, \beta] \}$$

$$\Rightarrow L(\Gamma_f) = \int_\alpha^\beta \sqrt{r^2(\varphi) + r'^2(\varphi)} d\varphi$$

$\Delta \circ \Gamma_f: \gamma(t) = \begin{cases} x = r \cos \varphi = f(\varphi) \cos \varphi \\ y = r \sin \varphi = f(\varphi) \sin \varphi \end{cases}$

$$\begin{aligned}
 \Rightarrow L(P_f) &= \int_a^b \sqrt{(f'(t) \cos t - f(t) \sin t)^2 +} \\
 &+ (f'(t) \sin t + f(t) \cos t)^2} dt = \\
 &= \int_a^b \sqrt{f'^2(t) + f^2(t)} dt = \\
 &= \int_a^b \sqrt{z^2(t) + z'^2(t)} dt \quad \triangle
 \end{aligned}$$

Необходимый интеграл

Опр Ф-ция f на E локально интегрируемой

$$f \in R_{loc}(E), \text{ если}$$

$$f \in R[a, b] \quad \forall [a, b] \subset E.$$

Опр. $\exists f \in R_{loc}(a, b), -\infty < a < b \leq +\infty$

$$\Rightarrow \text{символ } \int_a^b f dx$$

наз. несобств-м интегралом от f по $[a, b]$

Опр $\exists f \in R_{loc}(a, b), -\infty < a < b \leq +\infty$

$$\exists w \in [a, b]$$

$$\lim_{w \rightarrow b-0} \int_a^w f(x) dx, \text{ если } \exists b \in \mathbb{R} \text{ наз.}$$

значением н.и. от f по ин-ву $[a, b]$.

Опр $\exists f \in R_{loc}(a, b), -\infty < a < b \leq +\infty,$

то н.и. наз. сходящимся, если

$$\exists b \in \mathbb{R} \left[\lim_{w \rightarrow b-0} \int_a^w f dx \right], \text{ } b \text{ прот. символ}$$

Н. Ч. коз. *расходующимся.*

Tip. $\int_1^{+\infty} \frac{dx}{x^\alpha} = \lim_{b \rightarrow +\infty} \int_1^b \frac{dx}{x^\alpha} = \lim_{b \rightarrow +\infty} \begin{cases} \ln x \Big|_1^b, & \alpha = 1 \\ \frac{1}{(1-\alpha)x^{\alpha-1}} \Big|_1^b, & \alpha \neq 1 \end{cases}$

$$= \lim_{b \rightarrow +\infty} \begin{cases} \frac{\ln b}{(1-\alpha)b^{\alpha-1}} - \frac{1}{1-\alpha}, & \alpha \neq 1 \\ \frac{1}{\alpha-1}, & \alpha \leq 1 \end{cases}$$

$$\Rightarrow \text{cse. npxn} \quad \frac{\alpha > 1}{\text{P/cse npxn } \alpha \leq 1}$$
